Investigation of the Robustness of XAI-Based Federated Learning Against Adversarial Attacks for Smart Grid False Data Detection

Islam Elgarhy, Mahmoud M. Badr, Mohamed Mahmoud, Jianbing Ni, *Senior Member, IEEE*, Maazen Alsabaan, and Tariq Alshawi, *Senior Member, IEEE*

***Abstract*—Federated learning (FL) enables decentralized training of machine learning (ML) models, making it a valuable approach for detecting false data in smart power grids (SGs) to enhance grid stability while protecting consumers' privacy. However, FL-based ML models remain vulnerable to adversarial attacks during both training and inference phases, which can compromise data security. To address these vulnerabilities, we first investigate the robustness of a novel FL-based false data detection approach using explainable artificial intelligence (XAI), referred to as XAI-based FL detection. This approach utilizes explanations of consumers' power consumption data, rather than raw data, during the training process. We assess the robustness of the XAI-based FL detection compared to traditional data-driven FL detection against two types of adversarial attacks: 1) Gradient Inversion attacks in the training phase, where adversaries reconstruct private data from shared gradients, and 2) Evasion attacks in the inference phase, where adversaries subtly modify input data to deceive the detection model. Then, we propose a secure XAI-based FL detector with adversarial training (AT) to defend against both attack types. The key idea is that XAI helps mask model gradients during training because XAI-generated explanations remain nearly identical across different samples. Therefore, attackers struggle to accurately reconstruct the original training data, even if they obtain precise explanations using gradient inversion attacks. Additionally, XAI effectively distinguishes between benign and malicious samples. When combined with AT, XAI strengthens model robustness against evasion attacks without compromising accuracy, effectively resolving the tradeoff between security and performance. Our proposed detector reduced the success rate of evasion attacks from 94.99% to 29.11% with training on explanations, and further to 0% with adding AT. It also increased the mean square error for gradient inversion attacks from 0.01 to 2.60 in the most severe attack scenarios, making such attacks ineffective.**

***Index Terms*—Adversarial Attacks, explainable Artificial Intelligence (XAI), electricity theft, evasion, false data detection, federated learning (FL), gradient inversion, privacy preservation, smart grid.**

I. Introduction

SMART power grids (SGs) offer superior protection against physical attacks, such as meter tampering, compared to traditional grids. This is made possible through the use of advanced metering infrastructure (AMI), which facilitates bidirectional communication between utility companies and the smart meters (SMs) at consumer locations [1]. Despite this advantage, SGs are still exposed to cyber-attacks, such as false power consumption data injection. These attacks pose significant threats to SGs, causing significant disruptions to grid stability and energy management [2], [3]. Therefore, machine learning (ML) models are utilized to detect and mitigate such threats.

ML, trained using federated learning (FL), is crucial in SGs for ensuring energy efficiency and service reliability while preserving consumers' privacy [4]. FL enables the decentralized training of ML models, allowing multiple consumers, referred to as participants, to collaborate without sharing raw data. This approach effectively addresses privacy concerns associated with the centralized training of ML models [5], [6]. However, ML models have recently become targets of a growing class of attacks, known as adversarial attacks, which exploit weaknesses in these models during both the training and inference phases [7].

In the training phase, gradient inversion attacks pose significant privacy risks, where an adversary can invert the shared model gradients of FL participants to reconstruct the sensitive data [8], [9]. In case of fine grained power consumption readings, gradient inversion attacks could reveal personal information, such as daily routines, appliance usage, and presence at home, potentially leading to malicious

Received 28 April 2025; accepted 30 May 2025. Date of publication 3 June 2025; date of current version 25 July 2025. This work was supported by Ongoing Research Funding program, (ORF-2025-636), King Saud University, Riyadh, Saudi Arabia, and NSF grant number 2349104. *(Corresponding author: Mohamed Mahmoud.)*

Islam Elgarhy is with the School of Engineering and Technology, Western Carolina University, Cullowhee, NC 28723 USA, and also with the Department of Computer System, Faculty of Computer and Information Sciences, Ain Shams University, Cairo 11566, Egypt (e-mail: iaelgarhy42@tntech.edu).

Mahmoud M. Badr is with the Department of Network and Computer Security, College of Engineering, State University of New York Polytechnic Institute, Utica, NY 13504 USA, and also with the Department of Electrical Engineering, Faculty of Engineering at Shoubra, Benha University, Cairo 11672, Egypt (e-mail: badrm@sunypoly.edu).

Mohamed Mahmoud is with the Department of Electrical and Computer Engineering, Tennessee Tech University, Cookeville, TN 38505 USA (e-mail: mmahmoud@tntech.edu).

Jianbing Ni is with the Department of Electrical and Computer Engineering and Ingenuity Labs Research Institute, Smith Engineering, Queen's University, Kingston, ON K7L 3N6, Canada (e-mail: jianbing.ni@queensu.ca).

Maazen Alsabaan is with the Department of Computer Engineering, College of Computer and Information Sciences, King Saud University, Riyadh 11451, Saudi Arabia (e-mail: malsabaan@KSU.EDU.SA).

Tariq Alshawi is with the Department of Electrical Engineering, College of Engineering, King Saud University, Riyadh 11421, Saudi Arabia (e-mail: talshawi@ksu.edu.sa).

Digital Object Identifier 10.1109/JIOT.2025.3576225

2327-4662 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.

exploitation. Additionally, during the inference phase, an adversary exploits evasion attacks to subtly modify false data in a way that bypasses the ML model (i.e., causing incorrect decisions), with these modifications often being imperceptible to humans [10], [11].

The main challenge in defending against such attacks is the tradeoff between security and performance [12], [13]. Defense strategies often require sacrificing some level of the model's accuracy to enhance the model's robustness. Differential privacy (DP) is commonly used against gradient inversion attacks [14]. This technique introduces noise into the model's gradients, making it harder for attackers to reconstruct the original data. However, this added noise reduces the precision of the model's gradients, ultimately impacting its accuracy [15]. Additionally, encryption-based defenses ensure privacy protection by enabling computations on encrypted data. While they address the issue of sacrificing accuracy, they introduce a new challenge—high computational overhead [16], [17]. Similarly, adversarial training (AT) is commonly used against evasion attacks [18]. This involves training the model on adversarial samples—data that has been intentionally manipulated to mislead the model. While this helps the model better recognize and resist evasion tactics, it can also impair its ability to make accurate predictions on clean data. Therefore, *there is a critical need for a secure defense mechanism against adversarial attacks that not only enhances the model's robustness but also preserves its accuracy.*

On the other hand, explainable artificial intelligence (XAI) improves the transparency of ML models by offering clear insights into how decisions are made. It achieves this by generating explanations that are easy for humans to understand, such as highlighting the specific features or patterns in input data that have the most influence on the model's decisions [19], [20]. Recent research has established a connection between evasion attacks and XAI by detecting these attacks through anomalies in the model's explanations—specifically, deviations from the expected patterns observed in normal samples [21]. As a result, *XAI holds significant potential in securing ML models against adversarial threats*, offering the dual advantage of improving both model interpretability and performance.

The main goal of this work is to propose a method for securing an FL false data detector against adversarial attacks using XAI and AT as defense mechanisms. XAI plays a key role during the FL training process by training on explanations of power consumption data instead of raw data. This helps protect against gradient inversion attacks by obscuring the model's gradients, since XAI-generated explanations remain similar across different data samples, making it harder for attackers to reconstruct the original data. Additionally, XAI improves resilience against evasion attacks by clearly distinguishing between normal and abnormal power consumption patterns, which boosts AT's ability to detect such attacks while maintaining the model's accuracy. To the best of our knowledge, *this is the first work to propose a secure FL approach that leverages training with XAI to improve robustness while maintaining accuracy.*

The main contributions of our work are as follows.

1) *We have compared the robustness of both the XAI-based and existing Data-based FL false data detectors against two types of adversarial attacks targeting privacy and performance.*
   a) The comparison includes evaluations of various adversarial attacks: Deep leakage from gradients (DLG), Fast gradient sign method (FGSM), and Basic iterative method (BIM). These attacks were assessed in both independent and identically distributed (IID) and Non-IID settings.
2) *We propose a secure XAI-based FL false data detector with AT to enhance robustness while maintaining accuracy.*
   b) This detector is a deep ML model trained adversarially using XAI-generated explanations of power consumption data instead of the raw data, for both normal and adversarial samples. This approach enhances robustness and alleviates the tradeoff between security and performance.
3) *We have assessed our proposed secure XAI-based FL false data detector using a real-world electricity consumption dataset.*
   c) The experimental results demonstrate that the proposed detector achieves superior robustness while maintaining accuracy against DLG, FGSM, and BIM adversarial attacks, utilizing the Irish smart energy trial (ISET) dataset [22] under various levels of attacker capabilities.

The structure of this article is as follows. Section II reviews prior work on the use of FL in SGs security and the integration of XAI with adversarial ML. Section III describes the proposed robust XAI-based FL false data detector against adversarial attacks. Section IV presents the dataset, experimental setup, and evaluations. Finally, Section V concludes this article.

II. Related Works

In this section, we begin by reviewing the latest advancements in utilizing FL for SGs security. Next, we explore the connection between XAI and adversarial ML. Finally, we highlight the limitations and research gaps that this article aims to address.

A. FL in SGs Security

Several studies have explored the use of FL in securing SGs, improving grid performance and stability while preserving data privacy. These studies address the vulnerabilities of SGs to various cyber-attacks [6], [23], [24], [25], [26], [27], with some specifically focusing on adversarial attacks [28], [29], [30], [31].

Wen et al. [23] proposed a privacy-preserving FL framework for detecting energy theft in SGs. They utilize DP and homomorphic encryption to ensure data confidentiality. Their framework provides a secure and effective solution for enhancing SGs security, achieving high detection accuracy with lower computational overhead. Additionally, Li et al. [24] proposed a secure federated deep ML approach for detecting

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.

false data injection attacks in SGs, integrating the Transformer model, FL, and the Paillier cryptosystem to enhance detection accuracy while preserving data privacy. Their experimental results on IEEE 14-bus and 118-bus test systems confirm its effectiveness and superior performance.

Jithish et al. [6] proposed a privacy-preserving FL anomaly detection model for detecting abnormal behaviors in SGs, utilizing secure communication protocols for model training. Their model achieves performance comparable to centralized ML models while demonstrating efficiency in memory, CPU usage, bandwidth, and power consumption, making it suitable for deployment on resource-constrained edge devices. Similarly, Ibrahem et al. [25] proposed a privacy-preserving and efficient decentralized aggregation scheme based on a functional encryption cryptosystem, allowing detection stations to securely train a global FL model without exposing customers' private data. Their model effectively detects energy theft with low computational overhead.

Singh et al. [27] proposed a privacy-preserving FL model for predicting future energy requirements in SGs using serverless cloud computing and blockchain-enabled dew servers. Their model reduces cybersecurity risks, computation and communication costs, and attack probability while enhancing prediction accuracy and privacy preservation. Additionally, Badr et al. [26] proposed a privacy-preserving and communication-efficient FL energy predictor for net-metering systems. Their approach incorporates a hybrid deep ML model, functional encryption, and a change-and-transmit method. The predictor enables secure data aggregation and enhances communication efficiency while preserving customer privacy.

Unlike [6], [23], [24], [25], [26], [27], Tran et al. [28] proposed an efficient, privacy-preserving cross-silo FL approach for false data detection, addressing the vulnerability to inference adversarial attacks from dishonest aggregators. Their approach employs a two-layer encryption scheme, secret sharing, and parallel computing. The approach ensures privacy protection against honest-but-curious aggregators while achieving desirable model utility. Additionally, Bondok et al. [29] investigated the robustness of FL false data detector against adversarial evasion attacks. They found that AT can effectively defend against such attacks when FL participants are honest. However, its effectiveness diminishes when participants act maliciously.

Li et al. [30] proposed a privacy-preserving FL approach for SGs while addressing poisoning adversarial attacks. Their approach incorporates a dynamic adaptive defense mechanism to mitigate malicious gradients and identify malicious participants while ensuring privacy through multidimensional homomorphic encryption and hierarchical aggregation. The approach achieves superior performance and efficiently detects malicious participants. Additionally, Bondok et al. [31] proposed two defense mechanisms to counter poisoning trojan adversarial attacks in an FL false data detection approach. They assume that malicious participants in FL training manipulate their local models using specially crafted low-power-consumption data, which injects a trojan into the global model. The Trojan remains dormant until the inference phase, allowing attackers to bypass detection. Their evaluation highlights the strong capability of the proposed defenses against such attacks.

B. XAI and Adversarial ML

The integration of XAI with ML has attracted growing research interest. XAI techniques, such as local interpretable model-agnostic explanations (LIME) [32] and Shapley additive explanations (SHAP) [33], offer valuable insights into the reasoning behind the model's decisions, thereby enhancing reliability and trust. Moreover, there is a significant connection between XAI and adversarial ML, both in terms of attacks and defenses. Recent studies have explored the role of XAI in adversarial ML, with a focus on defense [34], [35], [36], [37], [38], [39], [40] and attack [41], [42] strategies.

Watson et al. [34] proposed a model-agnostic, explainability-based approach for accurately detecting adversarial samples in medical diagnosis datasets. Their approach improves adversarial detection by over 10% compared to the state-of-the-art and generalizes to various attack types without the need for retraining. Additionally, Fidel et al. [35] proposed a novel detection approach for evasion attacks in deep ML models. They utilized SHAP values computed for the internal layers of the model to distinguish between normal and adversarial samples across two benchmark image datasets. Their proposed detector demonstrates high detection accuracy and strong generalization ability in detecting various evasion attacks. Similarly, AL-Essa et al. [36] proposed an approach that combines AT and XAI to enhance the accuracy of deep ML models for cyber-threat detection, using FGSM to generate adversarial samples and SHAP for local explanations, which are then used for fine-tuning the model. Their experiments on two benchmark cybersecurity datasets demonstrate the effectiveness of the approach in improving classification accuracy of cyber-data.

Sandeepa et al. [37] proposed a robust aggregation approach that utilizes SHAP to identify potential poisoners in FL. Their approach employs feature attribution clustering, which mitigates poisoning attacks and provides improved explainability and justification for detecting and eliminating malicious clients during the aggregation process. Similarly, Gong et. al. [38] proposed a novel approach to enhance the robustness, fidelity, and efficiency of Byzantine-robust aggregation rules in FL. Their method improves the ability to distinguish between benign and malicious updates, effectively boosting detection capabilities. It also integrates seamlessly with existing Byzantine-robust mechanisms, resulting in significant performance gains across various datasets and against poisoning attacks.

Gholian et al. [39] proposed a novel explainability-driven approach to identify adversarial vulnerabilities in spatio-temporal mobile traffic forecasting models. Their approach leverages XAI, enabling network operators to better understand model behavior, detect security risks, and improve the robustness of forecasting models against adversarial threats. Additionally, Elgarhy et al. [40] proposed a robust and accurate false data anomaly detector, trained using XAI explanations generated by SHAP method. Their experimental results demonstrate that their approach is robust against

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.

various evasion attacks under a black-box threat model, while maintaining superior classification performance.

Unlike the use of XAI as a defense mechanism against adversarial attacks in [34], [35], [36], [37], [38], [39], and [40], Amich et al. [41] proposed a novel approach that utilizes feature-based explanations of ML model predictions to enhance adversarial sample generation. Their approach minimizes the perturbations of state-of-the-art attacks while improving the evasion rate. As a model-agnostic technique, it supports various threat models and distance metrics. The approach was successfully evaluated on two benchmark image datasets, demonstrating stable performance across multiple runs. Additionally, Zhang et al. [42] proposed a novel class of attacks that deceive both deep ML models and their interpretation methods. Their empirical evaluations show that these attacks enable an adversary to manipulate both model predictions and their corresponding explanations, posing significant risks, particularly in medical diagnosis systems.

C. Limitations and Motivations

As discussed in the previous sections, FL-based ML models are widely used to enhance the security of SGs. However, they remain vulnerable to adversarial threats. Moreover, there is a lack of research on incorporating XAI during FL training to enhance robustness and classification accuracy. Therefore, several limitations in FL-based ML models remain unaddressed, as outlined below.

1) *High Overhead and a Tradeoff Between Robustness and Accuracy:* FL-based models are susceptible to adversarial attacks, such as gradient inversion and evasion attacks. To counter these threats, defenses such as Encryption-based methods, DP, and AT are employed. However, encryption-based defenses impose high computational overhead while DP and AT defenses typically sacrifice model accuracy to improve robustness.
2) *Limited Exploration of Using XAI in FL for Adversarial Defense:* XAI is widely used to enhance the interpretability of ML models. Additionally, it can better differentiate between benign and anomalous data, thereby improving both accuracy and robustness. However, its potential for strengthening security in FL-based ML models has not been fully explored.

To address these challenges, this article explores a novel FL training approach that leverages XAI to enhance robustness while preserving accuracy. While most existing studies focus on using XAI to provide explanations in FL for better decision understanding, we propose a novel approach by adversarially training the FL-based false data detector on explanations rather than raw data to overcome the limitations discussed earlier. This approach is based on *the observation that XAI explanations for different samples may produce highly similar results*. This similarity complicates attackers' ability to identify specific patterns in the training data through gradient inversion attacks. Additionally, XAI can strengthen AT defense against evasion attacks because of its ability to distinguish between benign and malicious samples, thereby improving the model's robustness and maintaining its accuracy.

III. Robust XAI-Based FL False Data Detector Against Adversarial Attacks

In this section, we begin by outlining the different types of attacks addressed in this article and introducing the threat model considered. Then, we provide a detailed explanation of the proposed XAI-FL detector's architecture, incorporating relevant background concepts to explain the rationale behind its design.

A. Attacks and Threat Model

In this article, we consider false data attacks designed to falsify power consumption readings, along with two adversarial attacks—gradient inversion and evasion—that aim to either degrade model accuracy or compromise privacy. We focus on well-known and widely benchmarked adversarial attacks, including FGSM, BIM, and DLG, because they have established effectiveness in adversarial testing and provide meaningful insights into model robustness. These attacks require less prior knowledge from the attacker's side, making them more practical and posing a more significant threat in real-world scenarios. Furthermore, we investigate two threat model scenarios, each representing different attacker objectives and targeting distinct phases: 1) training and 2) inference.

1) False Data Attacks: Jokar et al. [43] proposed a false data attacks to simulate how an adversary manipulates actual (or benign) power consumption readings, R_b, specifically through a continuous attack. The continuous attack function reduces all power consumption readings R_b by a constant reduction factor α, where $(0 < \alpha < 1)$, as modeled by (1), to generate a false data sample R_m that is reported to the electrical utility

$$R_m = f(R_b) = \alpha \times R_b. \tag{1}$$

2) Evasion Attacks: The objective of evasion attacks is to manipulate a false data sample R_m, which is generated by the continuous attack function, by introducing small, deliberate changes (i.e., perturbations) to generate an evasion data sample. These changes are designed to mislead detectors, causing them to mistakenly classify the manipulated data as benign when it should have been flagged as malicious. Gradient-based evasion attacks leverage the gradient of the cost function with respect to the ML model's input. These attacks can be executed using single-step methods, such as the FGSM [44], or multistep methods such as the BIM [45]. Multistep methods iteratively refine the perturbations, creating more sophisticated and harder-to-detect adversarial samples, which increases the likelihood of misclassification. Both FGSM and BIM modify the input by following the sign (direction) of the gradient, aiming to maximize the classification loss for the true label, as modeled in (2) and (3), respectively

$$R_m = R_m + \epsilon.\ sign\Big(\nabla_{R_m} L_\theta(R_m, y, \theta)\Big) \tag{2}$$

$$R_{m_{i+1}} = clip_{(-\epsilon,+\epsilon)}\Big(R_{m_i} + \beta.\ sign\Big(\nabla_{R_{m_i}} L_\theta\big(R_{m_i}, y, \theta\big)\Big)\Big). \tag{3}$$

Here, the hyperparameters ϵ, β, and I define the perturbation magnitude, step size, and number of iterations, respectively.

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.

y represents the label of the input sample, θ refers to the parameters of the ML model, L_θ is the model's cost function, and ∇ denotes the gradient of the cost function with respect to θ.

3) Gradient Inversion Attacks: In [46], Zhu et. al. proposed a gradient inversion attack, called DLG, which threatens data privacy in FL by reconstructing training data samples from model gradients. DLG is an iterative, optimization-based approach that seeks to recover the original data by finding an input that generates gradients similar to those of the actual training samples. In practice, an adversary using the DLG attack minimizes the gradient matching distance D between the gradients of a randomly initialized data point (x, y) and those of the ground-truth data (x^*, y^*). This optimization process gradually refines (x, y), bringing it closer to the ground-truth data distribution (x^*, y^*), as modeled by

$$\arg\min_{(x^*,y^*)} D\big(\nabla_\theta L_\theta(x, y), \nabla_\theta L_\theta\big(x^*, y^*\big)\big). \tag{4}$$

Here, $\nabla_\theta L_\theta(x, y)$ and $\nabla_\theta L_\theta(x^*, y^*)$ represent the gradients ∇ of the loss function L with respect to ML model parameters θ. The distance D is used to iteratively update (x, y) until it closely resembles the original training samples. This is measured using the Euclidean L_2 norm between the actual (or true) gradients and the dummy gradients generated during reconstruction.

4) Threat Models: We consider two threat model scenarios. The focus in these scenarios is not only on the threat of false data attacks but also on the threat posed by adversarial attacks—either to compromise data privacy in Scenario *S#1* or to increase the likelihood of bypassing detection in Scenario *S#2*. The details of both scenarios are as follows.

1) *Scenario S#1:* It occurs during the training phase, where an attacker utilizes gradient inversion attacks to reconstruct training data from the gradients shared during the FL training. In this scenario, the attacker is the electric utility company (EUC), which functions as a centralized server in FL, aggregating local models to compute the global model. The EUC is considered *honest-but-curious*—it adheres to the training protocol as expected but may still analyze received gradients to infer sensitive customer data, potentially violating privacy. Although the literature suggests using cryptographic techniques to allow the EUC to aggregate the global model and defend against these attacks, these methods are not practical as they require substantial computational resources to perform all necessary operations on encrypted data. In both Data-FL and XAI-FL, the attacker employs the DLG adversarial attack to reconstruct the data from gradients. However, in XAI-FL, where gradients represent a model trained on XAI explanations, the attacker first applies DLG to extract these explanations from the gradients. Then, a multiregression model, trained on a historical public dataset, is used to recover power consumption data using the reconstructed XAI explanations.
2) *Scenario S#2:* It occurs during the inference phase, where attackers launch evasion attacks to bypass detection. In this scenario, the attacker is a malicious consumer who may either participate in the FL or not. If the malicious consumer does not participate in FL training, they lack access to both the training dataset and the model architecture. Consequently, they train a surrogate model with a different architecture and dataset, which may represent a single behavior as local data (referred to as *A#1*) or various behaviors as global data (referred to as *A#2*). If the malicious consumer does participate in the FL training, they gain access to the exact model and its local dataset (referred to as *A#3*). In both cases, attackers use FGSM and BIM evasion attacks to generate adversarial samples. These samples are then sent to the defense model in an attempt to bypass it. However, evasion samples generated in *A#3* tend to be more effective, as they are crafted using the exact model.

B. Rationale Behind the Proposed Detector

The proposed robust XAI-based FL detector integrates three key techniques: 1) FL; 2) XAI; and 3) AT. We first present the background of each technique individually before illustrating how their combination enhances both security and accuracy, ultimately contributing to a secure XAI-FL-based false data detector against adversarial attacks.

1) *FL:* It is a collaborative ML approach that enables multiple participants N to train a shared model while keeping their data private [47]. Each participant i trains a local model on its dataset W_t, generating gradient updates w_t^i. These updates are sent to a central server, which aggregates them to refine the global model, as modeled in (5). Here, W_t denotes the global model after t training rounds. The FL process terminates either upon reaching a predefined number of rounds or when the model achieves convergence

$$W_{t+1} = W_t + \frac{1}{n}\sum_{i=1}^{N} w_t^i. \tag{5}$$

2) *Shapley Additive Explanations (SHAP):* It is a XAI model-agnostic method introduced in [33] based on cooperative game theory. It surpasses alternative XAI methods, particularly in terms of consistent and stable feature importance, because of its foundation in Shapley values, which ensures robust and reliable attributions, leading to better alignment with human intuition [48], [49]. This method assigns importance scores to each input feature for ML model f using the Shapley value framework. The Shapley value for a feature i is calculated as the weighted average of its marginal contributions across all possible subsets S of the remaining features, as modeled in (6). Here, N represents the full set of features, $f(S)$ denotes the model's prediction using only features in subset S, and $f(S \cup \{i\})$ is the prediction when feature i is included. Practically, SHAP's original sampling-based approach is computationally expensive. Therefore, approximation-based approaches such as KernelSHAP and TreeSHAP are used to overcome this challenge. KernelSHAP is

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.

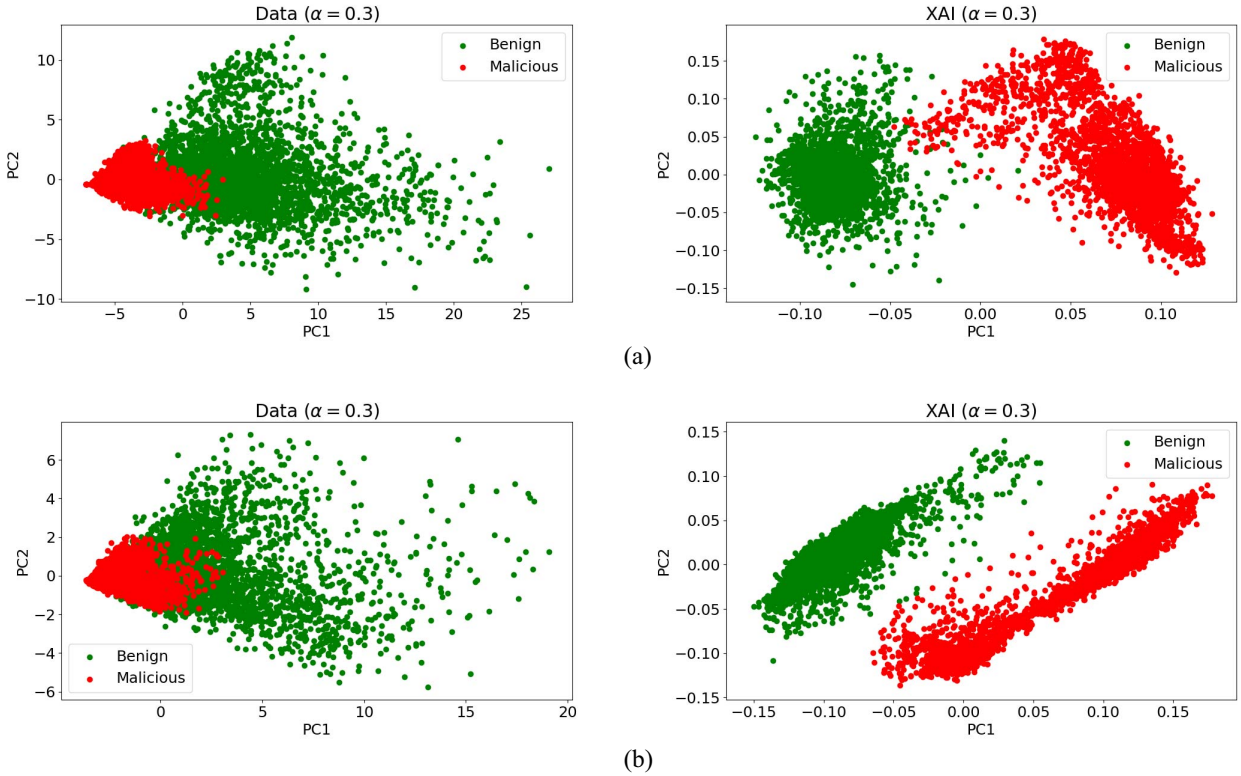


Fig. 1. PCA applied to Data and XAI explanations of power consumption readings. (a) IID. (b) Non-IID.

a model-agnostic method that balances the tradeoff between background samples and explanation fidelity. However, TreeSHAP, designed for tree-based models, derives SHAP values directly from the tree structure rather than relying on a background dataset

$$\text{SHAP}_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (f(S \cup \{i\}) - f(S)). \quad (6)$$

3) *Adversarial Training (AT):* This approach involves generating adversarial samples, labeling them as malicious, and incorporating them into the original dataset to strengthen the model's robustness against evasion attacks. The ML model is then trained on a combination of benign, malicious, and adversarial samples by simulating adversarial behavior, computing adversarial perturbations, and learning from these modified inputs [18]. This leads to the following min-max saddle point formulation for AT, as represented in (7). Here, θ represents the model parameters, $\mathcal{D}$ is the data distribution, $L_\theta(x, y)$ is the loss function for input x with label y, and δ is the adversarial perturbation constrained by a set $\mathcal{S}$ (e.g., $||\delta|| \leq \epsilon$ in l_p-norm). Instead of directly feeding samples from $\mathcal{D}$ into the loss function, the model first applies perturbations by solving the inner maximization problem to find the worst-case perturbation, while the outer minimization optimizes θ to improve robustness against such perturbations

$$\min_\theta \mathbb{E}_{(x,y)\ \sim \mathcal{D}} \left[\max_{\delta \in \mathcal{S}} L_\theta(x + \delta, y) \right]. \quad (7)$$

The proposed detector is trained through FL using XAI and AT. FL preserves privacy by averaging models across participants, as shown in (5). The XAI method, practically SHAP, utilizes feature importance, as shown in (6). SHAP enhances the distinction between normal and malicious samples. This can be demonstrated through principal component analysis (PCA) applied to both raw data and their corresponding XAI features in both IID and Non-IID settings. PCA reduces the dimensionality for better visualization. Fig. 1 shows the separation between benign and malicious data under the continuous attack described in (1). This separation emphasizes the improved performance and robustness that can be achieved by using XAI features instead of relying solely on raw data. Additionally, XAI-generated explanations remain somewhat similar across different data samples, which increases robustness by making it more difficult for attackers to reconstruct the original data when the model is trained on XAI features. Figs. 2 and 3 demonstrate this consistency for global and local explanations, respectively. In Fig. 2, data from two different clusters are used to display global explanations, while two samples are randomly chosen from within these clusters to illustrate local explanations in Fig. 3. These figures highlight that the top 10 features in the global explanations are similar, and the local explanations remain consistent, even though the samples come from different clusters representing different patterns. AT enhances the model's resilience to evasion attacks, as illustrated in (7). Unlike traditional AT, which often sacrifices accuracy for robustness, the use of XAI ensures that AT enhances robustness without significantly compromising accuracy. Therefore, integrating FL, XAI, and AT provides two

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.

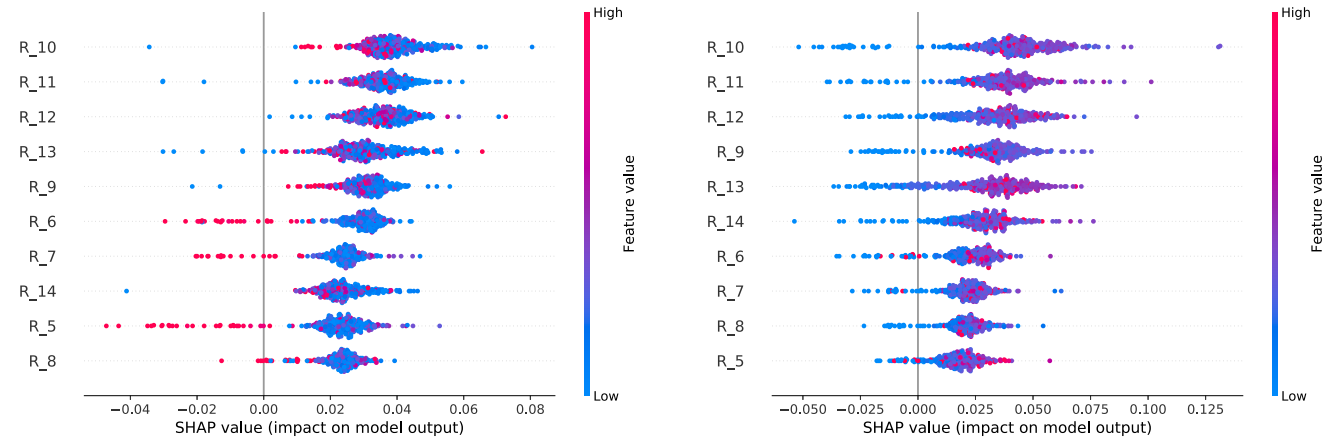


Fig. 2. SHAP global explanation of data from two different clusters (Non-IID).

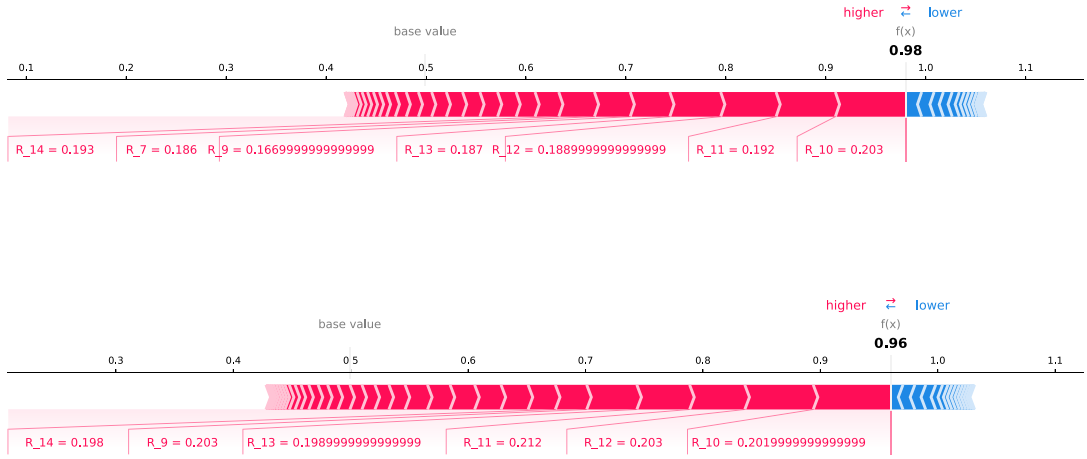


Fig. 3. SHAP local explanations of two random samples taken from two different clusters (Non-IID).

key advantages. First, XAI enhances both model performance and data privacy by limiting direct training on raw training data, thereby reducing the risk of data leakage from adversarial gradient inversion attacks. Second, the combination of XAI and AT strengthens the detection and mitigation of adversarial evasion attacks while maintaining high classification accuracy.

C. Proposed Detector Architecture

Fig. 4 illustrates the training and inference phases of the proposed XAI-based FL approach, which incorporates AT. Instead of independently training a false data detector by centralizing all data collected from SMs that periodically record fine-grained power consumption readings, the EUC collaborates with consumers to develop the detector using FL. This collaboration is necessary because consumers are unwilling to share their detailed power consumption data directly with the EUC, as it could compromise their privacy and expose sensitive information. Considering vulnerabilities to adversarial attacks—such as gradient inversion during the training phase and evasion attacks during the inference phase—the details of these phases are outlined below.

1) *Training Phase:* This phase involves N FL participants (i.e., consumers) collaborating with the EUC to train a global model over multiple FL communication rounds. Initially, each consumer generates explainable representations of its local data using a pretrained random forest model with the XAI SHAP method. Once these explanations are generated, consumers train their local models on them and participate in the FL process. Each consumer trains its model on adversarial samples using AT while periodically exchanging model updates with the EUC's aggregation server. Instead of sharing raw data, only model parameters are communicated, ensuring privacy preservation. This approach enhances the model's robustness against potential attacks, improving both security and performance.
2) *Inference Phase:* This phase involves communication between each consumer connected to the grid and the EUC. Here, each consumer i communicates with the

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.

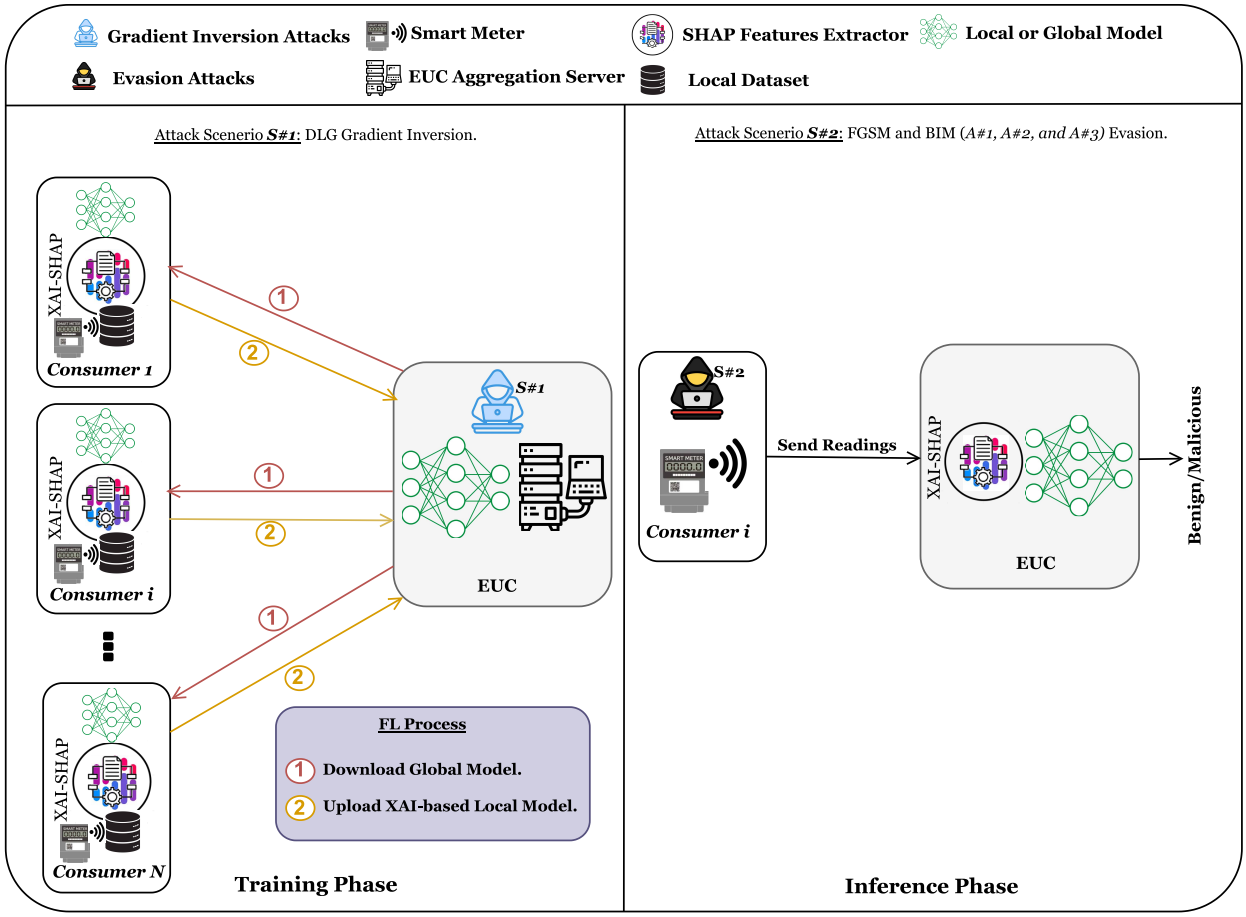


Fig. 4. Training and inference phases of the proposed detector.

EUC through the following three steps: First, the power consumption readings are sent from consumer's SM to the EUC. Second, the EUC extracts XAI representations (explanations) of these readings by applying SHAP to a pretrained model that was trained on a historical public dataset. Third, the EUC's global model classifies these readings as either benign or malicious.

IV. Performance Evaluation

In this section, we first introduce the dataset employed in our experiments, then describe the experimental setup and evaluation metrics. Lastly, we provide a detailed analysis of the experimental results.

A. Dataset

In this article, we utilize real electricity consumption data from the ISETs dataset [22], collected from over 3000 benign residential consumers (denoted as C). Each consumer's SM records a reading every 30 min, leading to 48 readings per day over a total of 536 days. However, the data collection process may introduce issues, such as incorrect readings, missing data, and outliers, often caused by equipment malfunctions, inaccuracies, or transmission errors. To ensure data integrity, we first apply data cleaning techniques to eliminate such errors [40]. The cleaned data is then used in our experiments, as will be explained in the following subsections. Additionally, to evaluate the proposed detector under both IID and Non-IID data settings, we apply the K-means algorithm to cluster consumers' power consumption data into distinct groups, with the number of clusters set to 11 using the elbow method. In the IID setting, all consumers are selected from a single cluster, ensuring similar consumption patterns. Conversely, in the Non-IID setting, one consumer is chosen from each cluster, capturing diverse consumption patterns that better reflect data variations. In both data settings, the dataset comprises 11 792 samples representing 11 consumers, where each consumer's benign and malicious data are combined and split into training and testing sets using a 2:1 ratio.

B. Experimental Setup and Evaluation Metrics

Our experiments were conducted on the high-performance computing cluster at Tennessee Technological University using Python libraries, including, pandas, Keras, Matplotlib, and scikit-learn. Additionally, the path-dependent TreeSHAP implementation [50], [51] was used as an approximation-based SHAP method, achieving fidelity in explanations with reduced computational overhead. To evaluate and compare the robustness of XAI-FL and Data-FL false data detectors, we used the dataset discussed in Subsection IV-A and applied various attacks discussed in Subsection III-A. To ensure a fair

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.

TABLE I
OPTIMAL HYPERPARAMETERS FOR CNN DETECTOR

Layer Type	Number of Units	Activation Function
Input	48	Linear
Conv1D	128	ReLU
Dense	256	ReLU
Dense	128	ReLU
Dense	64	Sigmoid
Output	2	Softmax

comparison, hyperparameters optimization was performed to determine the optimal architecture for both deep ML supervised detectors, practically the convolutional neural network (CNN) detector, and adversarial attacks.

CNN was chosen for both Data-based and XAI-based FL because of its superior ability to capture patterns and correlations in data through convolutional filters, making it highly effective in identifying both normal and abnormal patterns. The optimal architecture of the CNN detectors was determined through hyperparameter optimization, considering factors such as the number of layers, the number of units per layer, and the activation function (AF) used in each layer. Table I presents the optimal architecture for the CNN detector used during the FL process for both Data-based and XAI-based approaches. It begins with an Input layer containing 48 neurons with a linear AF. This is followed by a Conv1D layer with 128 neurons and ReLU AF. The architecture includes Dense layers with 256 and 128 neurons, both using ReLU AF, and a Dense layer with 64 neurons utilizing a Sigmoid AF. Finally, the output layer consists of 2 neurons with a Softmax AF, enabling binary classification (benign and malicious). Additionally, the Adam optimizer was used with 100 FL rounds, a learning rate of 0.0001, a batch size of 50, and Categorical Crossentropy as the loss function. The optimal hyperparameters for the adversarial attacks were determined by exploring a range of values and selecting those that most effectively maximize the attacker's success. For DLG attack, the optimal values includes the LBFGS optimizer with a learning rate of 0.001, Mean-Squared Error loss function, and 300 iterations. For FGSM and BIM attacks, the optimal values are as follows: β is a portion of the optimal ϵ, with $I = 100$ and $0.01 < \epsilon < 0.5$. Moreover, the optimal hyperparameters for the MultiOutputRegressor model include C = 1000, gamma = "auto," and epsilon = 0.05 for the support vector regression with kernel type = "rbf."

The following performance metrics, derived from (8) to (19), are used to evaluate the effectiveness of our FL detectors, as commonly done in the literature. The metrics include Accuracy (ACC), Detection Rate (DR), Precision (PR), F1-score ($F1$), False Alarm (FA), and Highest Difference (HD) for assessing the performance of detectors. The Attack Success Rate (ASR) is used to measure the effectiveness of evasion attacks. The Mean Absolute Error (MAE), Normalized MAE (*nMAE*), Mean-Square Error (MSE), Root MSE (RMSE), and Mean Arctangent Absolute Percentage Error (*MAAPE*) are used to quantify reconstruction accuracy of gradient inversion attacks. The ACC, DR, PR, $F1$, FA, and HD are derived from the confusion matrix, which includes True Positives (TP), True Negatives (TN), False Negatives (FN), and False Positives (FP). The ASR is calculated by comparing the DR before and after the attack, with a higher ASR indicating a more successful evasion attack, meaning it is more successful at evading the detector. The MAE, *nMAE*, *MSE*, RMSE, and *MAAPE* are calculated based on the difference between original and reconstructed power consumption readings, where lower values indicate better gradient inversion attack success, meaning the reconstruction closely matches the original data

$$\mathrm{ACC}(\%) = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \times 100 \tag{8}$$

$$\mathrm{DR}(\%) = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \times 100 \tag{9}$$

$$\mathrm{PR}(\%) = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \times 100 \tag{10}$$

$$F1(\%) = \frac{2 * \mathrm{DR} * \mathrm{PR}}{\mathrm{DR}+\mathrm{PR}} \times 100 \tag{11}$$

$$\mathrm{FA}(\%) = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}} \times 100 \tag{12}$$

$$\mathrm{HD}(\%) = \mathrm{DR} - \mathrm{FA} \times 100 \tag{13}$$

$$\mathrm{ASR}(\%) = \frac{\mathrm{DR}_{\mathrm{before}} - \mathrm{DR}_{\mathrm{after}}}{\mathrm{DR}_{\mathrm{before}}} \times 100 \tag{14}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |r_i - \hat{r}_i| \tag{15}$$

$$\mathrm{nMAE} = \frac{\sum_{i=1}^{n} |r_i - \hat{r}_i|}{\sum_{i=1}^{n} |r_i|} \times 100\% \tag{16}$$

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n} \left(r_i - \hat{r}_i\right)^2 \tag{17}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} \left(r_i - \hat{r}_i\right)^2} \tag{18}$$

$$\mathrm{MAAPE} = \frac{1}{n}\sum_{i=1}^{n} \arctan\left(\frac{|r_i - \hat{r}_i|}{|r_i|}\right) \times 100\%. \tag{19}$$

Here, TP refers to the number of malicious samples correctly classified, TN is the number of benign samples correctly classified, FN is the number of malicious samples incorrectly classified as benign, FP is the number of benign samples incorrectly classified as malicious, r_i and $\hat{r}_i$ are the original and reconstructed readings, and n is the total number of reading samples.

C. Experimental Results

We conducted two experiments to investigate the robustness of both the Data-based and XAI-based FL approaches against the two threat model scenarios described earlier in Section III-A. Experiment #1 was conducted to assess robustness against gradient inversion attacks in Scenario *S#1*, while Experiment #2 was conducted to assess robustness against evasion attacks in Scenario *S#2*, both under IID and Non-IID data settings. In both experiments, we trained Data-based FL (Data-FL) and XAI-based FL (XAI-FL) CNN detectors. The

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.

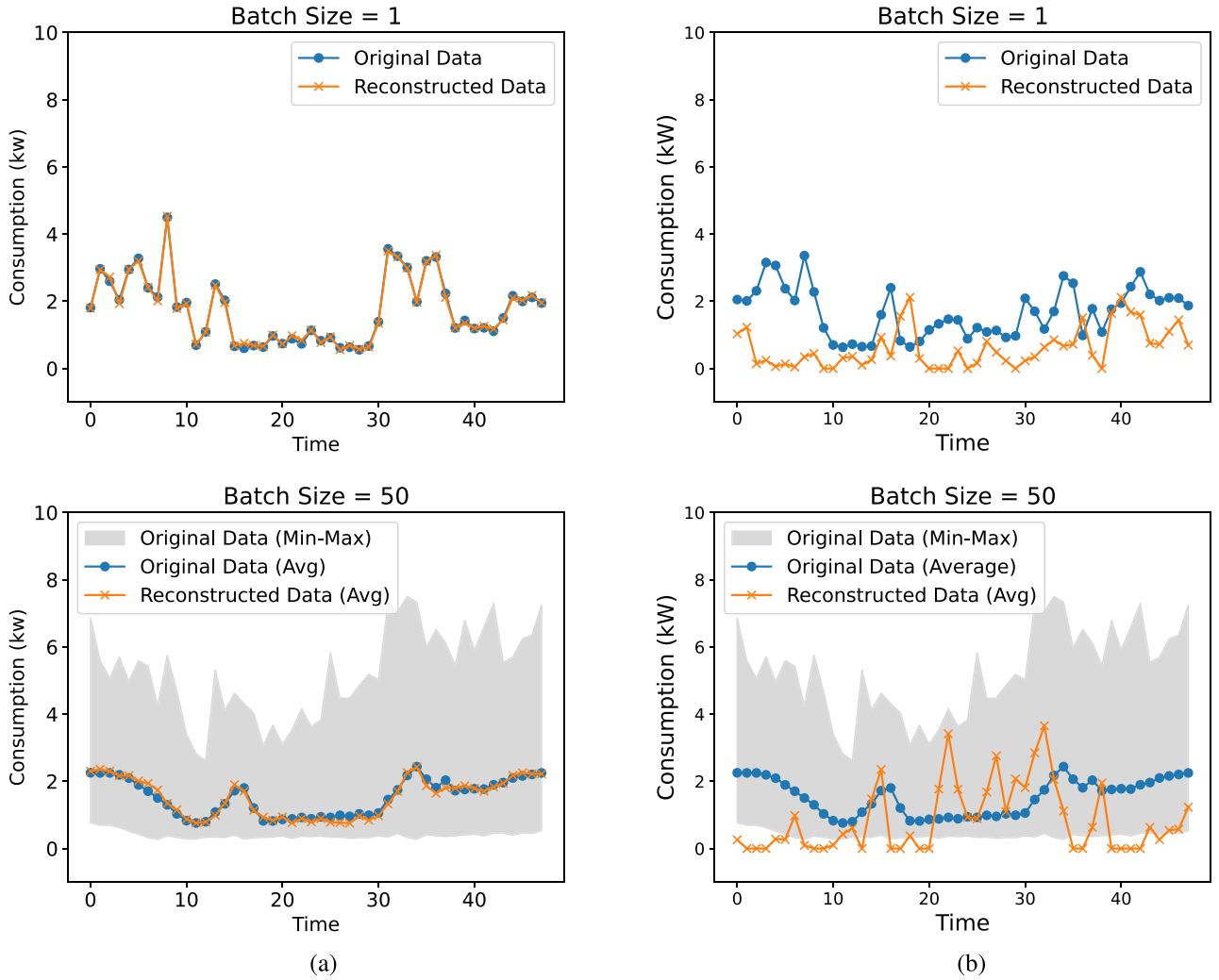


Fig. 5. Gradient Inversion Attacks against Data-FL and XAI-FL (IID setting) (a) Data-FL. (b) XAI-FL.

Data-FL detector is trained with raw power consumption data, while the XAI-FL detector is trained with XAI explanations of power consumption data. The details of these experiments are as follows.

1) Experiment #1 - Gradient Inversion Attacks: In this experiment, we evaluated the robustness of the two approaches (Data-FL and XAI-FL) against the gradient inversion DLG attack in terms of *MSE*, RMSE, MAE, *nMAE*, and *MAAPE* for both IID and Non-IID settings. The key results of this experiment are outlined as follows.

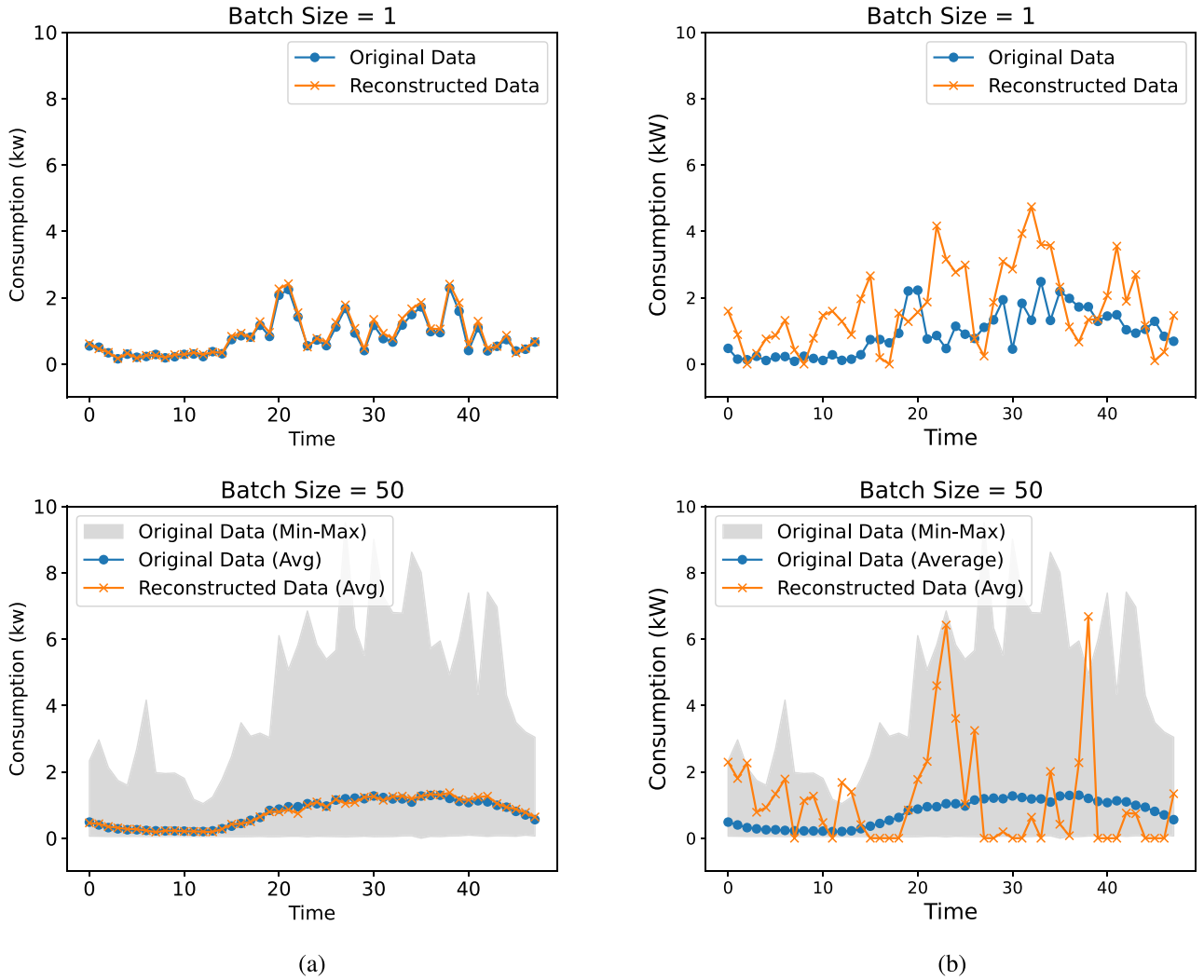
a) Fig. 5 provides a visual representation of the original and reconstructed data for both approaches (Data-FL and XAI-FL) using batch sizes of 1 and 50 in the IID setting. In Data-FL, it can be observed that the attacker can reconstruct the exact original data of a consumer when a batch size of 1, or approximate the average (Avg) original data when the batch size is 50, displaying the maximum (Max) and minimum (Min) power consumption readings for this consumer. However, in XAI-FL, the reconstructed data differs from the original data in both batch sizes (1 and 50). Additionally, Table II presents the reconstruction error

TABLE II
RECONSTRUCTION ERRORS OF GRADIENT INVERSION ATTACKS IN THE DATA-BASED AND XAI-BASED FL (IID SETTING)

	MSE	*RMSE*	*MAE*	*nMAE*	*MAAPE*
Data-FL	0.01	0.11	0.09	0.05	0.06
XAI-FL	1.94	1.39	1.21	0.79	0.62

performance of Data-FL and XAI-FL detectors against the DLG attack using a batch size of 50 for the IID setting. The metrics (*MSE*, RMSE, MAE, *nMAE*, and *MAAPE*) for Data-FL are lower than those for XAI-FL, indicating that Data-FL is more vulnerable to gradient inversion attacks, while XAI-FL is more robust against such attacks. For example, *MSE* and *MAAPE* are 0.01 and 0.06 for Data-FL, while for XAI-FL, they are 1.94 and 0.62, respectively. These results emphasize the severity of DLG privacy attacks and clearly demonstrate that the XAI-FL approach enhances robustness against such attacks. This enhancement is attributed to XAI's ability to mask the model's gradients, as XAI-generated explanations remain nearly identical across different data

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 6. Gradient Inversion Attacks against Data-FL and XAI-FL (Non-IID setting) (a) Data-FL. (b) XAI-FL.

samples, making it difficult for attackers to reconstruct the original training data. In contrast, when training on raw data, it is easier to reconstruct the data from the model's gradients because they directly reflect the original data.

b) Fig. 6 provides a visual representation of the original and reconstructed data for both approaches (Data-FL and XAI-FL) using batch sizes of 1 and 50 in the Non-IID setting. Similar to the IID setting, in Data-FL, the attacker can reconstruct the exact original data with a batch size of 1 or the average original data with a batch size of 50. However, in XAI-FL, the attacker struggles to reconstruct the original data in both batch sizes (1 and 50). Additionally, Table III presents the reconstruction error performance of Data-FL and XAI-FL detectors against the DLG attack using a batch size of 50 for the Non-IID setting. The metrics (*MSE*, RMSE, MAE, *nMAE*, and *MAAPE*) for Data-FL are lower than those for XAI-FL, indicating that XAI-FL is more robust against gradient inversion attacks. For example, *MSE* and *MAAPE* are 0.01 and 0.07 for Data-FL, while for XAI-FL, they are 2.60 and 0.89, respectively. These results clearly demonstrate that the XAI-FL approach enhances robustness against gradient inversion attacks because of the ability of XAI to mask the model's gradients.

TABLE III
RECONSTRUCTION ERRORS OF INVERSION ATTACKS IN THE DATA-BASED AND XAI-BASED FL (NON-IID SETTING)

	MSE	*RMSE*	*MAE*	*nMAE*	*MAAPE*
Data-FL	0.01	0.07	0.05	0.07	0.07
XAI-FL	2.60	1.61	1.18	1.53	0.89

2) Experiment #2 - Evasion Attacks: In this experiment, we evaluated the robustness of the two approaches (Data-FL and XAI-FL) against the FGSM and BIM evasion attacks in terms of classification performance, including ACC, PR, *F*1, DR, FA, and HD, and robustness against evasion attacks in terms of ASR for both IID and Non-IID settings. The key results of this experiment are outlined as follows.

a) Tables IV and V present the classification performance (ACC, PR, *F*1, DR, FA, and HD) of Data-FL and XAI-FL detectors without defense (No-Defense) and with AT

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
PERFORMANCE COMPARISON BETWEEN THE DATA-BASED AND XAI-BASED FL WITHOUT AND WITH AT DEFENSE (IID SETTING)

	No Defense		AT Defense	
	Data-FL	XAI-FL	Data-FL	XAI-FL
ACC	97.56	98.09	96.40	97.67
PR	98.90	98.64	93.28	96.09
$F1$	97.52	98.08	96.52	97.71
DR	97.71	97.53	100	99.38
FA	1.07	1.35	7.20	4.04
HD	96.64	96.18	92.80	95.34

TABLE V
PERFORMANCE COMPARISON BETWEEN THE DATA-BASED AND XAI-BASED FL WITHOUT AND WITH AT DEFENSE (NON-IID SETTING)

	No Defense		AT Defense	
	Data-FL	XAI-FL	Data-FL	XAI-FL
ACC	93.12	98.12	90.84	97.42
PR	94.39	96.62	84.52	95.13
$F1$	93.02	98.15	93.02	98.15
DR	91.69	99.72	100.00	99.94
FA	5.45	3.48	18.31	5.11
HD	86.24	96.24	81.69	94.83

defense for the IID and Non-IID settings, respectively. The results demonstrate that XAI-FL outperforms Data-FL in terms of classification performance, especially in the Non-IID setting. For example, in Table IV, under the No-Defense, ACC and FA are 97.56 and 1.07 for Data-FL, compared to 98.09 and 1.35 for XAI-FL. However, with AT Defense, XAI-FL preserves better classification performance compared to Data-FL, where ACC and FA are 96.40 and 7.20 for Data-FL, while for XAI-FL, they are 97.67 and 4.04, respectively. Similarly, in Table V, under the No-Defense column, ACC and FA are 93.12 and 5.45 for Data-FL, compared to 98.12 and 3.48 for XAI-FL. However, with AT Defense, XAI-FL preserves better classification performance compared to Data-FL, where ACC and FA are 90.84 and 18.31 for Data-FL, while for XAI-FL, they are 97.42 and 5.11, respectively. This is attributed to XAI's ability to distinctly separate benign from malicious consumption patterns as shown in Fig. 1, leading to *better performance and enhancing AT's ability to overcome its limitation of sacrificing model accuracy*.

b) Tables VI and VII compare the robustness (ASR) of Data-FL and XAI-FL detectors without defense (No-Defense) and with AT defense for the IID and Non-IID settings against evasion attacks *A*#1, *A*#2 and *A*#3 using FGSM and BIM. In the IID setting, the ASR values in Table VI demonstrate that XAI-FL is more robust against these attacks even without defense. For example, in *A*#3, the ASR for Data-FL is 84.81 and 90.11 for FGSM and BIM, respectively. However, for XAI-FL, these values are significantly reduced to 13.15 and 17.15. Additionally, both approaches achieve nearly similar robustness under AT defense, where ASR values are reduced to the same range. However, Data-FL sacrifices accuracy to achieve this robustness, whereas XAI-FL preserves accuracy due to the integration of XAI within AT, as shown in Table IV. Similarly, In the Non-IID setting, the ASR values in Table VII demonstrate that XAI-FL is more robust against these attacks even without defense. For example, in *A*#3, the ASR for Data-FL is 83.15 and 94.99 for FGSM and BIM, respectively. However, for XAI-FL, these values are significantly reduced to 15.20 and 29.11. Additionally, both approaches achieve nearly similar robustness under AT defense, where ASR values are reduced to the same range. However, Data-FL sacrifices accuracy to achieve this robustness, whereas XAI-FL preserves accuracy due to the integration of XAI into AT, as shown in Table V.

TABLE VI
ASR OF EVASION ATTACKS IN THE DATA-BASED AND XAI-BASED FL WITHOUT AND WITH AT DEFENSE IN (IID SETTING)

		No Defense		AT Defense	
		Data-FL	XAI-FL	Data-FL	XAI-FL
A#1	FGSM	35.03	19.77	4.52	3.22
	BIM	45.13	21.02	5.22	6.02
A#2	FGSM	54.24	16.95	3.52	4.74
	BIM	70.41	17.66	5.01	3.33
A#3	FGSM	84.81	13.15	0	0
	BIM	90.11	17.15	0	0

TABLE VII
ASR OF EVASION ATTACKS IN THE DATA-BASED AND XAI-BASED FL WITHOUT AND WITH AT DEFENSE (NON-IID SETTING)

		No Defense		AT Defense	
		Data-FL	XAI-FL	Data-FL	XAI-FL
A#1	FGSM	68.36	8.22	6.32	5.72
	BIM	75.03	18.22	8.32	5.72
A#2	FGSM	70.55	9.15	6.40	5.12
	BIM	86.44	23.20	9.40	5.12
A#3	FGSM	83.15	15.20	0	0
	BIM	94.99	29.11	0	0

V. CONCLUSION

This article investigates the security of FL false data detectors in SGs, with a particular focus on how adversarial attacks during both the training and inference phases can compromise data integrity, leak sensitive information, and degrade system performance. Specifically, we analyze and compare two distinct FL approaches: 1) Data-based FL, which is directly trained on raw power consumption data, and 2) XAI-based FL, which is trained on XAI-generated explanations of this data. The experiments evaluate the resilience of these approaches against three adversarial attacks—DLG, FGSM, and BIM—under different threat model scenarios. The results consistently demonstrate that XAI-based FL is significantly more robust than Data-based FL, as it effectively mitigates the risk of data

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.

reconstruction and evasion attacks while preserving high classification accuracy. This improvement is attributed to XAI's ability to obscure sensitive gradient information, making it more difficult for attackers to reconstruct the original training data through DLG gradient inversion attack. Additionally, XAI facilitates a clearer distinction of benign and malicious samples, leading to improved performance and maintaining accuracy when employing AT defense aganist FGSM and BIM evasion attacks. This study highlights the potential of XAI-based FL as a more secure, privacy-preserving, and reliable approach for detecting false data in SGs, ensuring both robustness and accuracy in adversarial environments.

REFERENCES

[1] S. Borenius, P. Kekolahti, H. Hämmäinen, M. Lehtonen, and P. Mähönen, "Novel industry architectures for connectivity solutions in the smart distribution grids," *IEEE Access*, vol. 11, pp. 68093–68112, 2023.

[2] Z. Zhang et al., "Vulnerability of machine learning approaches applied in IoT-based smart grid: A review," *IEEE Internet Things J.*, vol. 11, no. 11, pp. 18951–18975, Jun. 2024.

[3] S. H. Mohammed et al., "A review on the evaluation of feature selection using machine learning for cyber-attack detection in smart grid," *IEEE Access*, vol. 12, pp. 44023–44042, 2024.

[4] A. A. Elshazly, I. Elgarhy, M. Mahmoud, M. I. Ibrahem, and M. Alsabaan, "A privacy-preserving RL-based secure charging coordinator using efficient FL for smart grid home batteries," *Energies*, vol. 18, no. 4, p. 961, 2025.

[5] R. Zheng, A. Sumper, M. Aragüés-Peñalba, and S. Galceran-Arellano, "Advancing power system services with privacy-preserving federated learning techniques: A review," *IEEE Access*, vol. 12, pp. 76753–76780, 2024.

[6] J. Jithish, B. Alangot, N. Mahalingam, and K. S. Yeo, "Distributed anomaly detection in smart grids: A federated learning-based approach," *IEEE Access*, vol. 11, pp. 7157–7179, 2023.

[7] K. N. Kumar, C. K. Mohan, and L. R. Cenkeramaddi, "The impact of adversarial attacks on federated learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 5, pp. 2672–2691, May 2024.

[8] Y. Huang, S. Gupta, Z. Song, K. Li, and S. Arora, "Evaluating gradient inversion attacks and defenses in federated learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 7232–7241.

[9] A. Hatamizadeh, H. Yin, P. Molchanov, A. Myronenko, W. Li, and P. Dogra, "Do gradient inversion attacks make federated learning unsafe?" *IEEE Trans. Med. Imag.*, vol. 42, no. 7, pp. 2044–2056, Jul. 2023.

[10] I. Elgarhy, M. Badr, M. Mahmoud, M. Nabil, M. Alsabaan, and M. Ibrahem, "Securing smart grid false data detectors against white-box evasion attacks without sacrificing accuracy," *IEEE Internet Things J.*, vol. 11, no. 20, pp. 33873–33889, Oct. 2024.

[11] N. Bouacida and P. Mohapatra, "Vulnerabilities in federated learning," *IEEE Access*, vol. 9, pp. 63229–63249, 2021.

[12] H. Zhang, Y. Yu, J. Jiao, E. Xing, L. E. Ghaoui, and M. Jordan, "Theoretically principled trade-off between robustness and accuracy," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 7472–7482.

[13] A. E. Ouadrhiri and A. Abdelhadi, "Differential privacy for deep and federated learning: A survey," *IEEE Access*, vol. 10, pp. 22359–22380, 2022.

[14] K. Wei et al., "Federated learning with differential privacy: Algorithms and performance analysis," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 3454–3469, 2020.

[15] P. R. Ovi, E. Dey, N. Roy, and A. Gangopadhyay, "Mixed quantization enabled federated learning to tackle gradient inversion attacks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 5046–5054.

[16] Y. Aono et al., "Privacy-preserving deep learning via additively homomorphic encryption," *IEEE Trans. Inf. Forensics Security*, vol. 13, no. 5, pp. 1333–1345, May 2018.

[17] R. Aziz, S. Banerjee, S. Bouzefrane, and T. L. Vinh, "Exploring homomorphic encryption and differential privacy techniques towards secure federated learning paradigm," *Future Internet*, vol. 15, no. 9, p. 310, 2023.

[18] A. Mądry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," *Stat*, vol. 1050, no. 9, p. 12, 2017.

[19] Z. A. E. Houda, B. Brik, and L. Khoukhi, "Why should i trust your IDS? An explainable deep learning framework for intrusion detection systems in Internet of Things networks," *IEEE Open J. Commun. Soc.*, vol. 3, pp. 1164–1176, 2022.

[20] P. Barnard, N. Marchetti, and L. A. DaSilva, "Robust network intrusion detection through explainable artificial intelligence (XAI)," *IEEE Netw. Lett.*, vol. 4, no. 3, pp. 167–171, Sep. 2022.

[21] Y.-C. Lin and F. Yu, "DeepSHAP summary for adversarial example detection," in *Proc. IEEE/ACM Int. Workshop Deep Learn. Test. Testing Deep Learn. (DeepTest)*, 2023, pp. 17–24.

[22] Commission for Energy Regulation (CER). "CER smart metering project—Electricity customer Behaviour trial, 2009–2010 [dataset]." 2012. [Online]. Available: https://www.ucd.ie/issda/data/commissionforenergyregulationcer/

[23] M. Wen, R. Xie, K. Lu, L. Wang, and K. Zhang, "FedDetect: A novel privacy-preserving federated learning framework for energy theft detection in smart grid," *IEEE Internet Things J.*, vol. 9, no. 8, pp. 6069–6080, Apr. 2022.

[24] Y. Li, X. Wei, Y. Li, Z. Dong, and M. Shahidehpour, "Detection of false data injection attacks in smart grid: A secure federated deep learning approach," *IEEE Trans. Smart Grid*, vol. 13, no. 6, pp. 4862–4872, Nov. 2022.

[25] M. I. Ibrahem, M. Mahmoud, M. M. Fouda, B. M. ElHalawany, and W. Alasmary, "Privacy-preserving and efficient decentralized federated learning-based energy theft detector," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, 2022, pp. 287–292.

[26] M. M. Badr et al., "Privacy-preserving and communication-efficient energy prediction scheme based on federated learning for smart grids," *IEEE Internet Things J.*, vol. 10, no. 9, pp. 7719–7736, May 2023.

[27] P. Singh, M. Masud, M. S. Hossain, A. Kaur, G. Muhammad, and A. Ghoneim, "Privacy-preserving serverless computing using federated learning for smart grids," *IEEE Trans. Ind. Informat.*, vol. 18, no. 11, pp. 7843–7852, Nov. 2022.

[28] H.-Y. Tran, J. Hu, X. Yin, and H. R. Pota, "An efficient privacy-enhancing cross-silo federated learning and applications for false data injection attack detection in smart grids," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 2538–2552, 2023.

[29] A. H. Bondok, M. Mahmoud, M. M. Badr, M. M. Fouda, M. Abdallah, and M. Alsabaan, "Novel evasion attacks against adversarial training defense for smart grid federated learning," *IEEE Access*, vol. 11, pp. 112953–112972, 2023.

[30] X. Li, M. Wen, S. He, R. Lu, and L. Wang, "A privacy-preserving federated learning scheme against poisoning attacks in smart grid," *IEEE Internet Things J.*, vol. 11, no. 9, pp. 16805–16816, May 2024.

[31] A. Bondok, M. Badr, M. Mahmoud, M. Alsabaan, M. Fouda, M. Abdullah, "Securing one-class federated learning classifiers against trojan attacks in smart grid," *IEEE Internet Things J.*, vol. 12, no. 4, pp. 4006–4021, Feb. 2025.

[32] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should i trust you? Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Disc. Data Min.*, 2016, pp. 1135–1144.

[33] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 4765–4774.

[34] M. Watson and N. A. Moubayed, "Attack-agnostic adversarial detection on medical data using explainable machine learning," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, 2021, pp. 8180–8187.

[35] G. Fidel, R. Bitton, and A. Shabtai, "When explainability meets adversarial learning: Detecting adversarial examples using SHAP signatures," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, 2020, pp. 1–8.

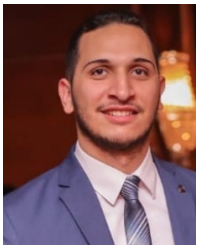
[36] M. Al-Essa, G. Andresini, A. Appice, and D. Malerba, "An XAI-based adversarial training approach for cyber-threat detection," in *Proc. IEEE Int. Conf. Depend. Auton. Secure Comput. Int. Conf. Pervasive Intell. Comput. Int. Conf. Cloud Big Data Comput. Int. Conf. Cyber Sci. Technol. Congr. (DASC/PiCom/CBDCom/CyberSciTech)*, 2022, pp. 1–8.

[37] C. Sandeepa, B. Siniarski, S. Wang, and M. Liyanage, "SHERPA: Explainable robust algorithms for privacy-preserved federated learning in future networks to defend against data poisoning attacks," in *Proc. IEEE Symp. Security Privacy (SP)*, 2024, pp. 204–204.

[38] Z. Gong et al., "AGRAMPLIFIER: Defending federated learning against poisoning attacks through local update amplification," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 1241–1250, 2023.

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.

[39] S. M. Gholian, C. Fiandrino, N. Vallina-Rodrìguez, M. Fiore, Widmer, and Joerg, "DeExp: Revealing model vulnerabilities for Spatio-temporal mobile traffic forecasting with explainable AI," *IEEE Trans. Mobile Comput.*, vol. 24, no. 6, pp. 5245–5263, Jun. 2025.
[40] I. Elgarhy, M. M. Badr, M. Mahmoud, M. Alsabaan, T. Alshawi, and M. Alsaqhan, "XAI-based accurate anomaly detector that is robust against black-box evasion attacks for the smart grid," *Appl. Sci.*, vol. 14, no. 21, p. 9897, 2024.
[41] A. Amich and B. Eshete, "EG-booster: Explanation-guided booster of ML evasion attacks," in *Proc. 12th ACM Conf. Data Appl. Security Privacy (CODASPY)*, 2022, pp. 16–28. [Online]. Available: https://doi.org/10.1145/3508398.3511510
[42] X. Zhang, N. Wang, H. Shen, S. Ji, X. Luo, and T. Wang, "Interpretable deep learning under fire," in *Proc. 29th USENIX Security Symp. (USENIX Security)*, 2020, pp. 1–9.
[43] P. Jokar, N. Arianpoo, and V. C. Leung, "Electricity theft detection in AMI using customers' consumption patterns," *IEEE Trans. Smart Grid*, vol. 7, no. 1, pp. 216–226, Jan. 2016.
[44] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.
[45] A. Kurakin, I. J. Goodfellow, and S. Bengio, *Adversarial Examples in the Physical World*. Hoboken, NJ, USA: CRC Press, 2018.
[46] L. Zhu, Z. Liu, and S. Han, "Deep leakage from gradients," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–7.
[47] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Artif. Intell. Stat.*, 2017, pp. 1273–1282.
[48] X. Man and E. Chan, "The best way to select features? Comparing MDA, LIME, and SHAP," *J. Financial Data Sci. Winter*, vol. 3, no. 1, pp. 127–139, 2021.
[49] C. Xu, Z. Liao, C. Li, X. Zhou, and R. Xie, "Review on interpretable machine learning in smart grid," *Energies*, vol. 15, no. 12, p. 4427, 2022.
[50] S. M. Lundberg et al., "From local explanations to global understanding with explainable AI for trees," *Nat. Mach. Intell.*, vol. 2, no. 1, pp. 56–67, 2020.
[51] A. Yasodhara, A. Asgarian, D. Huang, and P. Sobhani, "On the trustworthiness of tree ensemble explainability methods," in *Proc. Mach. Learn. Knowl. 5th IFIP*, Aug. 2021, pp. 293–308.



Islam Elgarhy received the B.S. and M.S. degrees in computer science from the Faculty of Computer and Information Sciences, Ain Shams University, Cairo, Egypt, in 2011 and 2019, respectively, and the Ph.D. degree in electrical and computer engineering from Tennessee Tech University, Cookeville, TN, USA, in 2025.

He is currently an Assistant Professor with the School of Engineering and Technology, Western Carolina University, Cullowhee, NC, USA. His research interests include machine learning, cryptography, network security, smart grids, explainable AI, and high performance computing.

Dr. Elgarhy has been selected as a poster winner in Tennessee Tech University's annual research and creative inquiry day, 2022.



Mahmoud M. Badr received the B.S. and M.S. degrees in electrical engineering (electronics and communications) from Benha University, Cairo, Egypt, in 2013 and 2018, respectively, and the Ph.D. degree in electrical and computer engineering from Tennessee Tech University, Cookeville, TN, USA, in 2022.

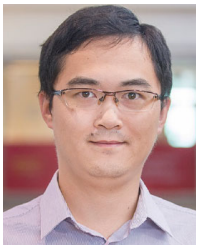
He is currently an Assistant Professor with the Network and Computer Security, Cybersecurity Department, College of Engineering, State University of New York Polytechnic Institute, Utica, NY, USA. He is also holding the position of Assistant Professor with the Faculty of Engineering at Shoubra, Benha University. His research interests include machine learning, blockchain, cryptography, 5G networks, and smart grids.



Mohamed Mahmoud received the Ph.D. degree from the University of Waterloo, Waterloo, ON, Canada, in April 2011.

He is currently a Professor with the Department of Electrical and Computer Engineering, Tennessee Tech University, Cookeville, TN, USA. His research interests include robust machine learning models and security and privacy-preserving schemes for smart grids.

Dr. Mahmoud is a recipient of the NSERC Postdoctoral Fellowship (Canada) and the Fulbright Award (USA). He has received Best Paper Awards at three IEEE international conferences and has authored over 150 publications in IEEE journals and conferences. He serves as an Associate Editor for the IEEE INTERNET OF THINGS JOURNAL and *Peer-to-Peer Networking and Applications* (Springer), and has been a technical program committee member for numerous IEEE conferences.



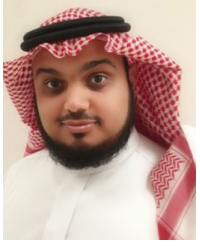
Jianbing Ni (Senior Member, IEEE) received the Ph.D. degree in electrical and computer engineering from the University of Waterloo, Waterloo, ON, Canada, in 2018.

He is currently an Assistant Professor with the Department of Electrical and Computer Engineering, Queen's University, Kingston, ON, Canada. His research interests are applied cryptography and network security, with current focus on edge computing, robust artificial intelligence, network security, and Blockchain technology.



Maazen Alsabaan received the B.S. degree in electrical engineering from King Saud University (KSU), Riyadh, Saudi Arabia, in 2004, and the M.A.Sc. and Ph.D. degrees in electrical and computer engineering from the University of Waterloo, Waterloo, ON, Canada, in 2007 and 2013, respectively.

He is currently an Associate Professor with the Department of Computer Engineering, KSU. From 2015 to 2018, he was the Chairperson of the Department. He serves as a consultant for different agencies and has been awarded many grants from KSU and King Abdulaziz City for Science and Technology, Riyadh. His current research interests include wireless communications and networking, surveillance systems, vehicular networks, green communications, intelligent transportation systems, and cybersecurity



Tariq Alshawi (Senior Member, IEEE) received the B.Sc. degree from King Saud University, Riyadh, Saudi Arabia, in 2009, the M.S. degree from the University of Michigan at Ann Arbor, Ann Arbor, MI, USA, in 2013, and the Ph.D. degree from Georgia Institute of Technology, Atlanta, GA, USA, in 2018.

He is a Faculty with the Electrical Engineering Department, King Saud University, Riyadh, Saudi Arabia. He has over 15 years of teaching and research experience in multidimensional signal processing, computer vision, and machine learning. He was the emerging technology advisor to his excellency the governor of Saudi telecom regulator, a Researcher with the Prince Sultan Advanced Technologies Research Institute, and the Center of Excellency in Optics & Radio at King Saud University.

Authorized licensed use limited to: North South University. Downloaded on October 30,2026 at 17:07:16 UTC from IEEE Xplore. Restrictions apply.